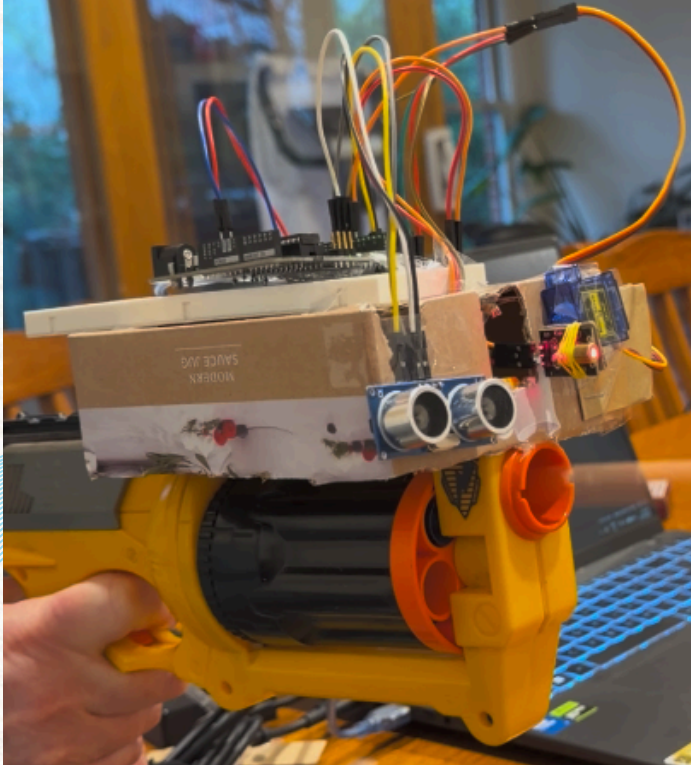


NERF GUN PROJECT

Personal project - Charlie Truscott

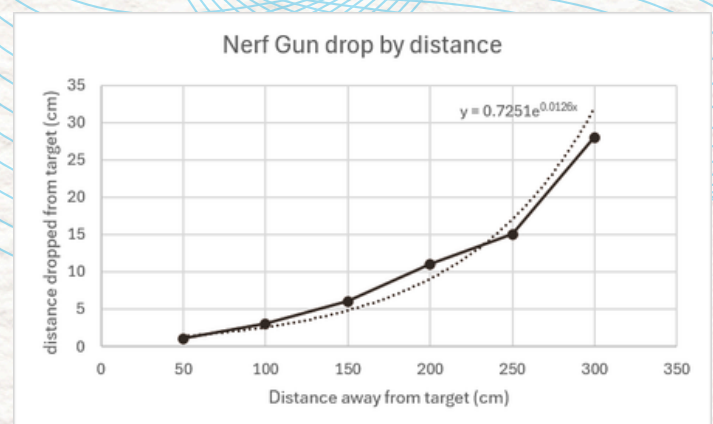


Trajectory Testing

Field-tested dart drop in 50 cm increments across the blaster's effective range (0.5–3 m) under controlled conditions gun fixed at 90°, leveled, and stationary relative to the target, with distances measured to ± 0.5 mm accuracy. Fit the resulting data to an exponential drop model ($\text{drop} = 0.7251 \cdot e^{0.0126x}$), which implicitly accounts for air resistance. So now with the distance reading we can convert it into a servo correction angle via arctangent closing the loop between empirical testing and real-time aim correction.

Project Overview

Designed a laser-sighting system for a Nerf blaster that automatically compensates for projectile drop. The system estimates dart trajectory based on target distance, using empirically derived drag and drop coefficients obtained through field testing of the darts' flight characteristics. A servo-actuated laser adjusts its angle in real time to match the calculated trajectory, giving the shooter an accurate point-of-aim regardless of range effectively an "aim-assist" system for the blaster.



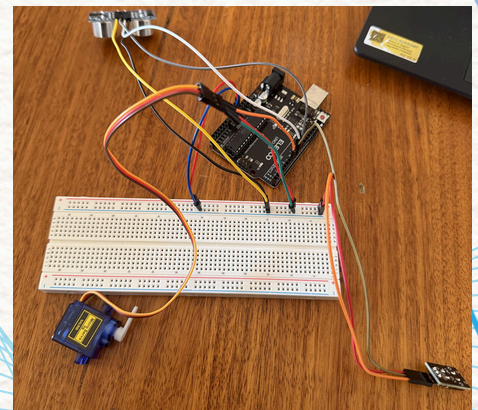
NERF GUN PROJECT

Simulation and Circuit design

For circuit design, the ultrasonic sensor, servo, and Arduino were wired and prototyped a simulation before finalizing the build. After the simulations were accurately running, i designed it against a physical breadboard with the arduino elements with the correct correction-angle calculation code to begin testing.

Final code:

The code implements distance-sensing LED controller using an ultrasonic sensor. The sensor to emit a pulse, measures the time for the echo to return, and converts that duration into a distance reading in centimeters using the speed of sound. With the brightness of the LED getting brighter the further away from the target. Each cycle, the distance and corresponding brightness are printed to the serial monitor for live monitoring.



Final product:

